

# SHAIMAA DWEDAR

## Software Engineer — Full-Stack Development

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### PROFESSIONAL SUMMARY

Computer Engineering graduate with hands-on, self-built experience across full-stack software development, backend systems, and AI/ML integration. Delivered end-to-end software independently — from REST APIs and MQTT-based backends to ML pipelines and cross-platform mobile apps — for both academic and freelance/client projects. Comfortable owning a project from architecture and code to a working, tested product.

### TECHNICAL SKILLS

**Languages:** JavaScript, Python, Dart

**Backend & Databases:** Node.js, Express.js, FastAPI, PostgreSQL, REST APIs, JWT auth

**Frontend:** HTML, CSS, JavaScript, React, Next.js, Three.js, responsive & bilingual (RTL) design

**AI / Machine Learning:** PyTorch, Hugging Face Transformers (BERT), scikit-learn, OpenAI GPT API integration

**Mobile Development:** Flutter, Dart, MVVM architecture, Firebase/Firestore

**IoT & Embedded:** ESP32, MQTT (incl. TLS), sensor/actuator control, firmware development, finite-state machines

**Tools & Practices:** Git/GitHub, basic Docker, MLflow, Kubeflow, AWS; audit logging, role-based access control

### EXPERIENCE

#### Freelance Software Developer — Self-employed / Remote

2026 – Present

- Deliver full-stack and IoT projects independently for clients, including a secure Node.js/Express backend controlling an ESP32-connected inverter over MQTT with role-based access, audit logs, and heartbeat monitoring.
- Build cross-platform mobile apps in Flutter/Dart, covering UI implementation, REST API/backend integration, bug fixing, and performance optimization, managing each project's lifecycle independently.

#### Software Development Trainee — Dimensions Info-tech, Ramallah

Summer 2025

- Developed features in Flutter using MVVM architecture, working against a PostgreSQL-backed system.
- Collaborated within a structured development workflow, applying professional coding and architecture practices in a team setting.

### KEY PROJECTS

#### Pizza Go — IoT Smart Pizza Ordering System (Graduation Project)

- Designed and built an end-to-end IoT ordering system: programmed two ESP32 controllers with embedded firmware to drive linear actuators, stepper motors, relay modules, a conveyor belt, and a sauce pump, backed by Node.js, PostgreSQL, MQTT, and Stripe for payments.
- Coordinated the full production workflow with a finite-state machine and heartbeat-based device monitoring; carried out hardware/firmware debugging, communication troubleshooting, and end-to-end testing and validation.

#### IdeasTracker — Voice-and-AI Idea Capture App

- Built an app that captures ideas by voice and organizes them with AI, using the Web Speech API for voice input, the OpenAI GPT API for processing, and Firebase for storage.

#### Inverter Control Backend (Freelance, with Power, for Rift Valley)

- Built a secure backend to control an ESP32-connected inverter over MQTT (TLS), with JWT authentication in HttpOnly cookies, role-based access control, audit logs, ACK validation, and command timeout handling.

#### Where Should I Go — AI Recommendation App

- Built a place-recommendation app using an LLM with tool-calling (OpenRouter) integrated with the Google Places API, plus a Flutter mobile version that recommends places based on mood, available time, and location.

### EDUCATION

#### B.Sc. Computer Engineering — Al-Quds University, Palestine

Sep 2021 – Jun 2026

### CERTIFICATIONS & ACHIEVEMENTS

- Front-End Web Development — Udacity (2024)
- Python and Machine Learning — The Hope (2022)
- IEEE Xtreme Programming Competition 2024 — Ranked 3rd at university level (Al-Quds University)